Errors can also originate from the client. We will look at what happens when a client makes a request for a page that doesn't exist, for example. You will learn how to use middleware—StatusCodePages middleware—to manage how your application responds to this very common event and how to control what the user sees.

Unless your users report back to you, you may never know that an error has taken place. Often, when a user does report a fault, they cannot remember precisely what they were doing, so you spend a lot of time trying to replicate the issue but lack the knowledge required to do so successfully. Ideally, you would like someone to watch over the user's shoulder who can report every detail of what they were doing when something went wrong. Clearly, that's not feasible, but the next best thing is including a mechanism to record user actions at the time an error occurred. This mechanism is called *logging*, and we will explore how it is supported within an ASP.NET Core application. You will learn how to categorize the information the logs capture, so that you can more easily control what is recorded, and we will explore the options for controlling where log output goes.

The default support for logging in ASP.NET Core is limited in terms of where logs are written. In the event you want to output logs to a different medium, you may need to implement your own log provider. This chapter covers the basic steps required to implement a simple provider that writes logs to email. Alternatively, rather than write and maintain your own logging solution, you might prefer to use a tried and trusted solution from a third party. There are plenty available that support ASP.NET Core applications. See http://mng.bz/Qngv for more details. You could use any of the solutions listed in the documentation, but in this book, I will walk you through installing and using one called Serilog.

Logging isn't just useful for error reporting. It can be used for tracking user behavior and auditing activity as well. The standard output format for most